

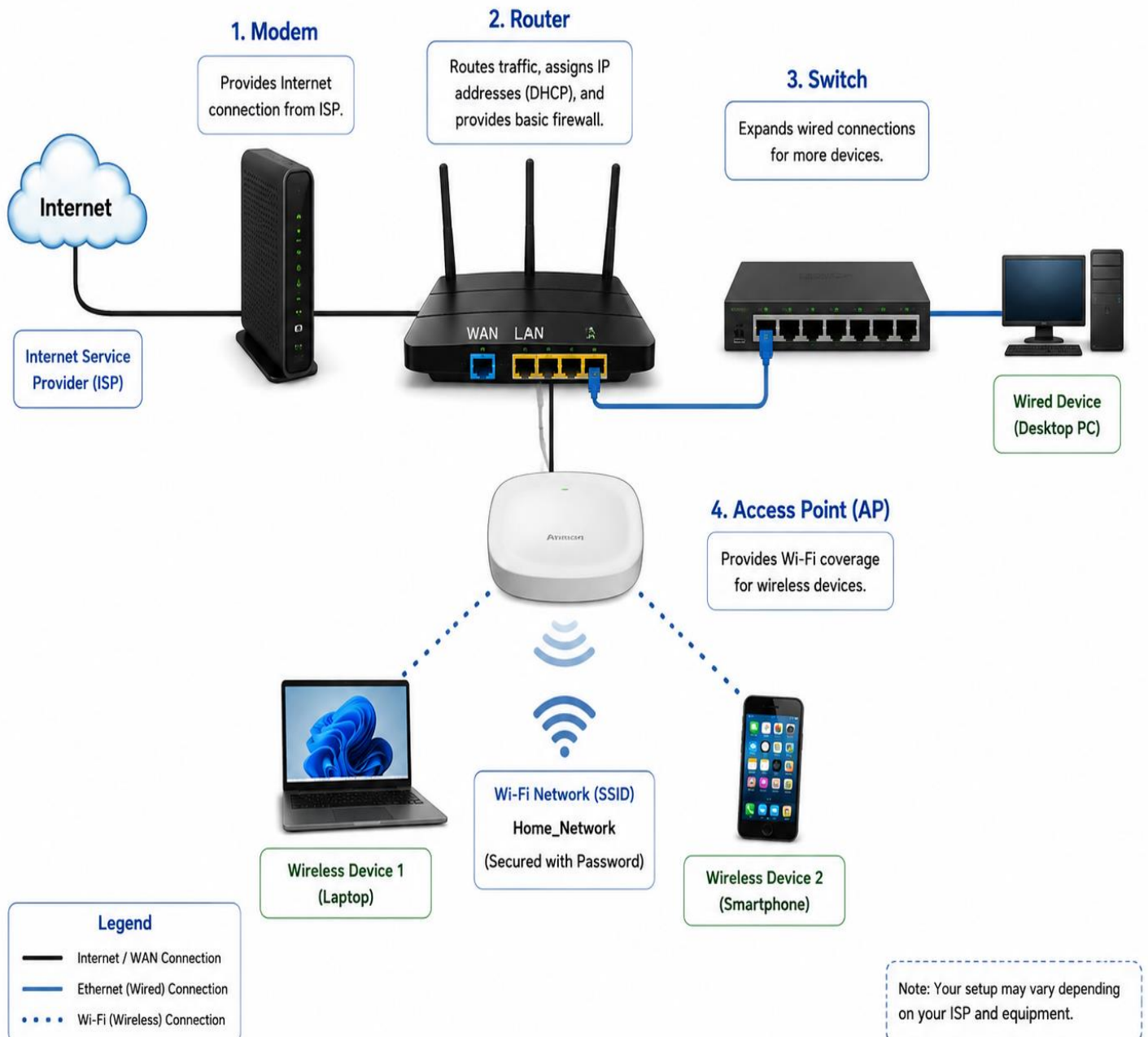
8/24/2026

Assignment-1

Create by
~pintu mer

- I. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

Typical Home Wi-Fi Setup

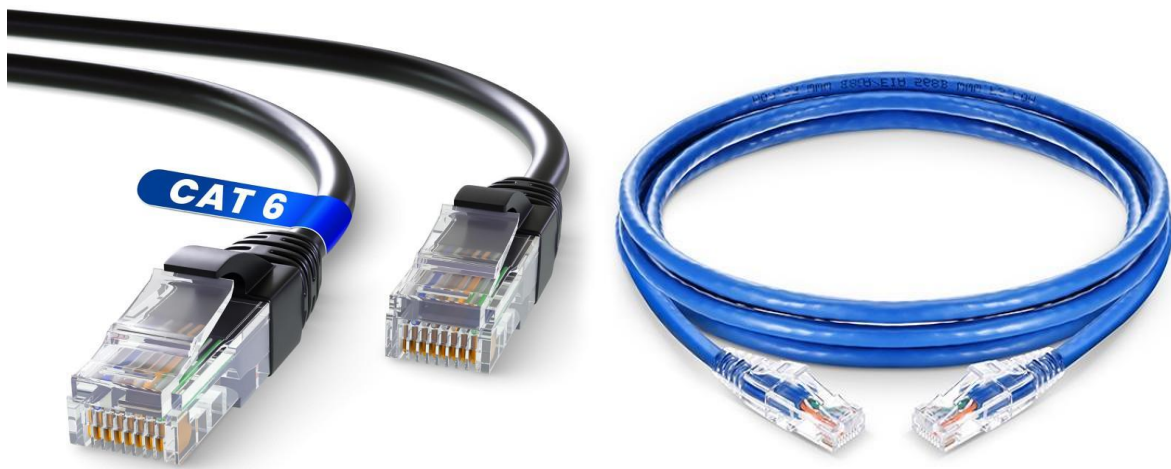


2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Networking Device	Main Function	Real-Life Example
Modem	Connects your home/network to the ISP (Internet Service Provider).	At home, the modem receives the internet connection from your ISP.
Router	Sends data between networks and shares the internet with multiple devices.	At home, a router connects your laptop, phone, and TV to the internet.
Switch	Connects multiple wired devices in the same local network (LAN).	In an office, a switch connects PCs, printers, and servers using Ethernet cables.
Access Point (AP)	Provides Wi-Fi wireless access to devices.	In a school or office, an AP provides Wi-Fi for laptops and smartphones.
Network Cable (Ethernet cable)	Carries network data between wired devices.	A LAN cable connects a desktop PC to a router or switch.

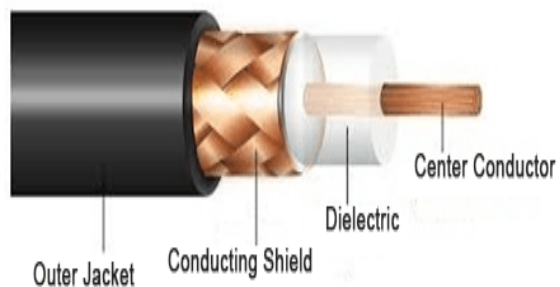
3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

1. Ethernet Cable (RJ45 / Cat6)



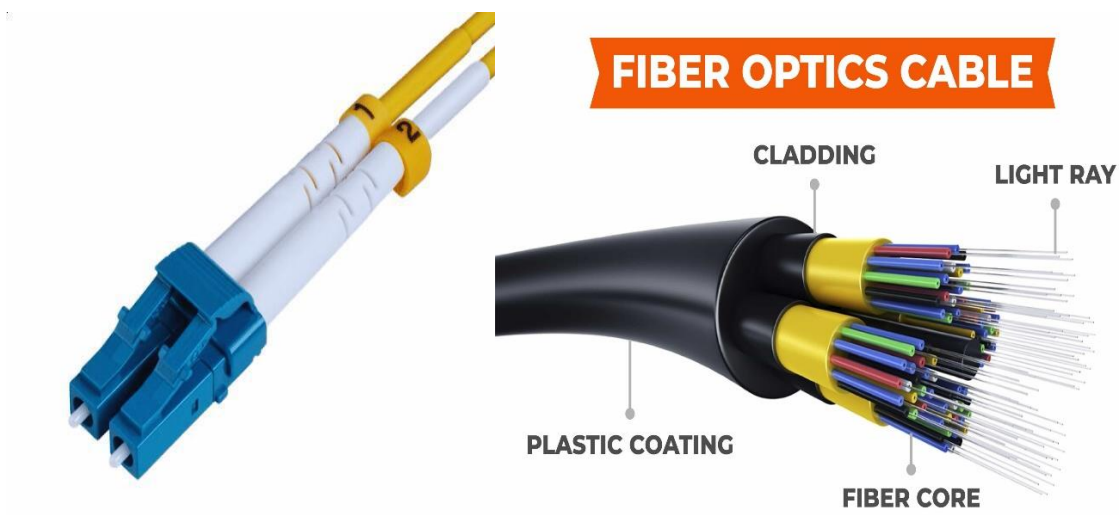
Common use: Used to connect computers, routers, switches, printers, and other devices in home/office LAN networks

2. Coaxial Cable



Common use: Commonly used for Cable TV, cable internet, CCTV, and antenna connections.

3. Fiber Optic Cable



Common use: Used for high-speed internet, data centers, telecom networks, and long-distance communication.

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	Router	Switch
Main Use	Connects different networks and provides internet access.	Connects devices within the same network.
Typical Location	Usually found between the modem/Internet and the local network, such as at home or office.	Usually found inside a LAN, such as an office or server room.

How it handles data	Uses IP addresses to decide where to send data.	Uses MAC addresses to forward data to the correct device.
Data Unit	Mainly handles packets between networks.	Mainly forwards Ethernet frames within a LAN.
Example	Home Wi-Fi router connects your home network to the Internet.	Office switch connects PCs, printers, and other devices together.
OSI Layer	Mainly Layer 3 (Network Layer).	Mainly Layer 2 (Data Link Layer).